

VLANs — Guided Notes ANSWER KEY

Unit 2.1.2 | CompTIA Network+ N10-009 Obj. 2.1 | N10-008 2.1

For instructor use / distribute after students complete the notes.

Question 1

A VLAN creates a separate _____ domain on a switch. Devices on different VLANs cannot communicate at Layer 2 even if they are plugged into the _____. VLAN IDs range from _____ to _____. Best practice is to avoid using VLAN _____ for production traffic.

Answer: Broadcast; same physical switch; 1; 4094; VLAN 1.

Question 2

An access port belongs to _____ VLAN and connects to _____. A trunk port carries traffic for _____ VLANs simultaneously and typically connects _____.

Answer: One; end devices (workstations, printers, cameras); multiple; switches to switches or switches to routers.

Question 3

802.1Q tagging inserts a _____ -byte tag into the Ethernet frame on trunk links. The tag contains the _____. When a frame exits an _____ port to an end device, the tag is _____.

Answer: 4-byte; VLAN ID; access; removed/stripped.

Question 4

The native VLAN is sent across a trunk _____. If the native VLAN doesn't match on both ends of a trunk, frames end up _____. Best practice is to change the native VLAN away from _____ and ensure it _____ on both sides.

Answer: Untagged (no 802.1Q tag); in the wrong VLAN (traffic leak/mismatch); VLAN 1; matches.

Question 5

Router-on-a-stick uses _____ on a single router interface — one per VLAN — each configured with the _____ IP for that VLAN. Its limitation is _____. Layer 3 switching uses _____ (SVIs) on the switch itself and routes at _____.

Answer: Subinterfaces; gateway; all inter-VLAN traffic funnels through one physical interface (bottleneck); Switched Virtual Interfaces; wire speed in hardware.

Question 6

A voice VLAN allows one physical port to carry both _____ traffic (untagged, data VLAN) and _____ traffic (tagged, voice VLAN). The IP phone negotiates its VLAN via _____ or _____. QoS policies on the voice VLAN ensure _____.

Answer: PC/data; voice; CDP; LLDP; voice packets are prioritized for call quality.

Question 7

Match each VLAN misconfiguration to its symptom:

Wrong VLAN assignment → _____

Trunk not formed → _____

VLAN not allowed on trunk → _____

No inter-VLAN routing → _____

Answer: Wrong VLAN → device gets wrong DHCP address or none, can't reach resources; Trunk not formed → VLANs don't extend between switches; VLAN not allowed → traffic for that specific VLAN won't cross the trunk; No routing → devices on different VLANs can't communicate even with correct addresses.

Question 8 — Real World Application

REAL WORLD: A new teacher's workstation is connected but can only reach devices in the front office, not classroom resources. Other teachers on the same switch can reach everything. What do you check first and why?

Answer: Check the VLAN assignment on that specific switch port with 'show vlan brief' or 'show interfaces switchport'. The port is likely assigned to the administration VLAN (front office) instead of the staff/classroom VLAN. Verify what IP address the workstation received — if it's in the admin subnet, that confirms the wrong VLAN. Correct the access VLAN assignment on the port and have the workstation renew its DHCP lease.